

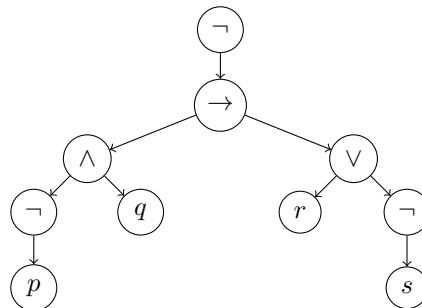
This exam has 3 pages and 8 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

Please motivate all answers!

1. Parse tree (5+6 points)

- (a) For the following parse tree, find the propositional logic formula that it represents.



- (b) Draw how the truth value of this formula is determined by means of its parse tree in a bottom-up fashion, if $p = \text{F}$, $q = \text{F}$, $r = \text{F}$, and $s = \text{T}$.

2. Disjunctive normal form (3+6 points)

- (a) Construct the truth table for the formula $(p \vee q) \rightarrow \neg r$.
- (b) Extract from this truth table a formula in DNF that is semantically equivalent to $(p \vee q) \rightarrow \neg r$.

3. OBDDs (3+8 points)

- (a) Represent $x \wedge (y \vee z)$ by means of a binary decision tree, with respect to the variable ordering x, y, z .
- (b) Reduce this binary decision tree to an ordered binary decision diagram. (Also give the intermediate reduction steps.)

4. Predicate logic (7+7 points)

For each of the following pairs of formulas, either argue that they are semantically equivalent, or give a model on which their truth values are different. In cases where semantic equivalence does not hold, explain moreover whether one of the formulas semantically entails the other.

(a) $\exists x (C(x) \vee D(x))$ and $(\exists x C(x)) \vee (\exists x D(x))$

(b) $\exists x (C(x) \wedge D(x))$ and $(\exists x C(x)) \wedge (\exists x D(x))$

5. Sets (6 + 6 points)

- (a) Suppose we have three sets A , B and C in a universe U , about which we have the following information:

$$\#A = 10, \quad \#B = 7, \quad \#C = 10,$$

$$\#(A \cap B) = 6, \quad \#(B \cap C) = 5, \quad \#(C \cap A) = 7,$$

$$\#(A \cap B \cap C) = 4, \quad \#((A \cup B \cup C)') = 5.$$

Use Venn diagrams to determine $\#(A \cup B \cup C)$ and $\#U$.

- (b) Using the algebra of sets, prove that the following set-theoretic equality holds for all sets A , B and C :

$$A \cap (B' \cup (B \cap C')) = (A' \cup (B \cap C))'.$$

6. Relations and functions (3 + 4 + 4 points)

In the set $V := \{1, 2, 3, 4, 5\}$, consider the binary relations R and S defined by

$$R := \{\langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 4, 1 \rangle, \langle 5, 4 \rangle\},$$

$$S := \{\langle 2, 1 \rangle, \langle 2, 3 \rangle, \langle 3, 4 \rangle, \langle 4, 2 \rangle, \langle 4, 5 \rangle\}.$$

- (a) Draw the directed graph representations of R and S .
- (b) For each of the two relations R and S , decide whether or not it is a function (from V to V). Please explain your answers.
- (c) Determine the relation $S^{-1} \circ R$, and give an explicit listing of its elements.

7. Ordering relations (8 + 3 points)

Let R and S be the partial ordering relations on the respective sets $A := \{a, b, c\}$ and $B := \{1, 2, 3\}$, specified by the following Hasse diagrams:



Consider the product set

$$A \times B = \{\langle a, 1 \rangle, \langle a, 2 \rangle, \langle a, 3 \rangle, \langle b, 1 \rangle, \langle b, 2 \rangle, \langle b, 3 \rangle, \langle c, 1 \rangle, \langle c, 2 \rangle, \langle c, 3 \rangle\}.$$

We are interested in the lexicographic order on $A \times B$ induced by the partial orders R on A , and S on B .

- (a) Use the algorithm you have learned to construct the Hasse diagram of the lexicographic order on the set $A \times B$: write down the sets G_x and H_x obtained in the construction, and use them to draw the Hasse diagram. (If you prefer, you can simplify the notation by writing **a1** for $\langle a, 1 \rangle$, **a2** for $\langle a, 2 \rangle$, and so on.)
- (b) Does the set $A \times B$ have a largest element according to your Hasse diagram? If so, please identify the largest element; if not, please list all maximal elements.

8. Induction (3 + 8 points)

Consider the sequence $(t_n)_{n=1}^{\infty}$ of integer numbers defined recursively by

$$t_1 := 0, \quad t_{n+1} = t_n + n(3n + 1).$$

We claim that for all $n \geq 1$,

$$t_n = n^2(n - 1).$$

- (a) Verify by explicit calculation that the claim is true for $n = 1$, $n = 2$, and $n = 3$.
- (b) Prove by mathematical induction that the claim is true for *all* integers $n \geq 1$.